

Numerical Linear Algebra First-Year Exam, May 2022

Do 4 (four) problems.

1. Prove or provide a counterexample to each of the following.

- (a) If matrix A is normal and triangular, then it is diagonal.
- (b) Every matrix has a Schur factorization.

2. This problem has two parts.

- (a) For $v \in \mathbb{C}^n$, define $f_A(v) := (v^*Av)^{1/2}$. Under what condition(s) on $A \in \mathbb{C}^{n \times n}$ is $f_A(\cdot)$ a norm on $\mathbb{C}^n$? (Full credit will only be given for a general condition or conditions. An example is not sufficient.) Prove that $f_A(\cdot)$ is a norm on $\mathbb{C}^n$ under the condition(s) you require.
- (b) Assuming the condition(s) you require in (a), given a matrix A , determine the best constants α and β to satisfy the inequality

$$\alpha\|v\|_2 \leq f_A(v) \leq \beta\|v\|_2.$$

3. Let $A = U\Sigma V^*$ be the singular value decomposition of $A \in \mathbb{C}^{m \times n}$. Let u_j denote column j of U .

- (a) Suppose $\text{rank}(A) = p < n < m$. Show $\{u_1, u_2, \dots, u_p\}$ is a basis for $\text{Col}(A)$, and $\{u_{p+1}, u_{p+2}, \dots, u_m\}$ is a basis for $\text{Null}(A^*)$.
- (b) Suppose A is full rank and $x \neq 0$. Let $\sigma_i, i = 1, \dots, n$, be the singular values of A . Show

$$\sigma_1 \geq \frac{\|Ax\|_2}{\|x\|_2} \geq \sigma_n > 0.$$

If you want to use the property that $\|A\|_2 = \sigma_1$, then you must prove that also.

4. Let $\{a_1, \dots, a_n\}$ be a linearly independent set of vectors. Consider the Gram–Schmidt and modified Gram–Schmidt algorithms for computing an orthonormal basis $\{q_1, \dots, q_n\}$ so that $\text{Span}\{q_1, \dots, q_k\} = \text{Span}\{a_1, \dots, a_k\}$, for each $k = 1, \dots, n$. Suppose in computing q_2 , an orthogonalization error is committed so that $q_2^*q_1 = \epsilon$.

- (a) Use the Gram–Schmidt algorithm to compute v_3 so that $q_3 = v_3/\|v_3\|$. What is $v_3^*q_2$?
- (b) Use the modified Gram–Schmidt algorithm to compute v_3 so that $q_3 = v_3/\|v_3\|$. What is $v_3^*q_2$?

5. Compute the Cholesky decomposition of the following matrix, or explain why it does not exist.

$$A = \begin{pmatrix} 1 & 1/2 & 2 & 3 \\ 1/2 & 5/16 & 3/2 & 5/2 \\ 2 & 3/2 & 17 & 17 \\ 3 & 5/2 & 17 & 31 \end{pmatrix}.$$